

SAN ANTONIO STINSON, TX, USA

WMO: 722523

Lat: 29.339N

Lon: 98.472W

Elev: 174

StdP: 99.25

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 12970

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.6	1.1	-11.2	1.5	8.3	-8.9	1.8	9.7	9.5	13.4	8.6	14.2	2.3	10	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.9	38.5	23.3	37.5	23.4	36.4	23.3	26.0	31.8	25.6	31.4	25.2	31.0	4.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	20.1	27.6	24.1	19.4	27.1	23.8	19.1	27.1	81.4	32.2	79.2	31.8	77.7	30.9	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.5	7.7	7.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-4.1	39.9	1.7	1.7	-5.3	41.2	-6.3	42.2	-7.3	43.1	-8.6	44.4		
			-5.5	26.5	1.5	0.6	-6.6	26.9	-7.5	27.3	-8.3	27.6	-9.5	28.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.7	12.0	14.6	18.1	21.9	25.6	28.8	29.6	30.2	27.5	22.6	17.0	12.6
	DBStd	7.53	4.88	5.48	4.52	3.70	3.02	2.00	1.87	1.78	2.76	4.29	4.95	4.85
	HDD10.0	91	35	18	3	1	0	0	0	0	0	1	6	27
	HDD18.3	688	202	127	61	12	1	0	0	0	0	14	83	188
	CDD10.0	4379	98	146	256	358	483	564	609	627	524	391	216	108
	CDD18.3	1933	6	22	54	119	226	314	351	369	274	147	42	11
	CDH23.3	20963	52	150	358	947	2159	3657	4293	4742	2958	1309	281	58
CDH26.7	10182	5	31	85	321	924	1866	2319	2663	1432	478	54	5	
Wind	WSAvg	3.1	2.9	3.2	3.5	3.7	3.8	3.6	3.2	2.9	2.6	2.7	2.7	2.7
Precipitation	PrecAvg	790	47	36	51	66	103	81	68	55	82	86	56	43
	PrecMax	1223	137	109	180	170	239	287	298	219	242	309	233	157
	PrecMin	320	3	6	0	6	16	9	0	0	0	0	0	7
	PrecStd	279	36	33	39	49	64	64	82	71	67	89	66	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.2	30.1	31.3	34.9	36.7	39.0	39.4	39.7	38.7	34.3	30.5	27.2
		MCWB	16.5	17.1	19.0	21.3	23.4	23.6	23.3	23.0	22.4	22.8	21.4	19.0
	2%	DB	24.8	27.2	28.9	32.1	34.6	37.3	38.1	38.5	36.6	32.6	28.1	25.0
		MCWB	16.0	17.7	18.7	20.2	22.9	23.9	23.5	23.1	22.7	22.5	19.7	17.7
	5%	DB	22.6	25.1	27.2	30.0	33.0	35.9	37.1	37.5	34.9	31.1	26.6	23.0
		MCWB	15.7	16.9	18.9	20.3	23.0	23.5	23.5	23.3	23.0	21.7	19.2	17.5
	10%	DB	20.8	22.8	25.2	28.3	31.5	34.2	35.8	36.4	33.3	29.6	24.7	21.2
		MCWB	15.4	16.7	17.7	20.0	22.7	23.5	23.6	23.3	23.0	21.2	18.9	16.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.4	21.3	21.9	24.4	25.8	26.7	26.6	26.4	26.1	25.7	23.2	21.4
		MCDB	22.7	25.4	26.9	30.1	31.9	33.2	32.6	31.0	31.0	29.7	26.6	24.5
	2%	WB	19.4	20.3	21.1	23.2	25.1	25.9	25.7	25.7	25.4	24.7	22.3	20.4
		MCDB	22.1	23.7	25.2	28.4	31.1	32.2	31.8	31.4	30.4	28.6	25.6	23.1
	5%	WB	18.0	19.5	20.5	22.5	24.4	25.4	25.3	25.2	25.0	24.0	21.3	19.4
		MCDB	20.8	22.6	24.2	27.1	29.8	31.3	31.3	31.4	30.0	27.9	24.6	21.7
	10%	WB	16.5	18.5	19.9	21.8	23.9	25.0	24.9	24.8	24.5	23.3	20.3	17.7
		MCDB	19.7	21.5	23.4	26.0	28.9	30.6	30.7	30.8	29.4	27.2	23.6	20.2
Mean Daily Temperature Range	5% DB	MDBR	12.2	11.9	11.6	11.5	10.9	10.9	11.3	11.9	11.0	11.7	11.7	11.9
		MCDBR	15.4	14.0	13.0	13.6	12.4	12.8	13.5	13.5	12.9	11.9	12.5	14.4
		MCWBR	8.7	7.4	6.1	5.6	3.8	2.7	2.7	2.7	3.1	4.3	6.2	8.4
	5% WB	MCDBR	11.3	10.1	10.2	9.8	10.2	10.2	10.8	11.2	10.1	9.4	9.6	10.6
		MCWBR	7.9	6.7	5.7	4.9	3.9	3.1	2.7	2.6	3.1	4.4	5.9	7.8
Clear-Sky Solar Irradiance	taub	0.331	0.337	0.361	0.404	0.451	0.432	0.439	0.424	0.417	0.375	0.363	0.321	
	taud	2.488	2.476	2.421	2.317	2.233	2.340	2.353	2.378	2.350	2.453	2.423	2.529	
	Ebn at Noon	907	932	926	892	847	858	851	863	859	878	861	902	
	Edh at Noon	92	101	113	130	142	127	125	121	120	102	97	84	
All-Sky Solar Radiation	RadAvg	3.02	3.55	4.49	5.33	5.77	6.60	6.64	6.36	5.22	4.34	3.26	2.68	
	RadStd	0.34	0.64	0.47	0.54	0.59	0.42	0.58	0.36	0.52	0.57	0.36	0.35	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	+0.18	N/A	N/A	N/A	+254	+188
Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+107	+90

Nomenclature: See separate page